

Ujjwal Jha

Hyderabad, India

+91 7260059459

2001ujjwaljha@gmail.com — [LinkedIn](#) — [GitHub](#) — [Portfolio](#)

PROFESSIONAL SUMMARY

Generative AI & Full-Stack Engineer with 3 years of experience building production-grade AI systems and scalable distributed backend architectures. Specialized in Agentic AI systems, Retrieval-Augmented Generation (RAG), LLM orchestration, and high-integrity transactional systems. Strong expertise in concurrency control, database design, asynchronous processing, and cloud-native deployments.

TECHNICAL SKILLS

- **Generative AI & Agents:** LangChain, LangGraph, CrewAI, OpenAI API, RAG pipelines, hybrid retrieval, embeddings, prompt engineering, hallucination mitigation, context optimization
- **Backend Engineering:** Python, FastAPI, Node.js, NestJS, Spring Boot, REST APIs, OpenAPI, microservices architecture
- **Databases:** PostgreSQL, Oracle XE, MongoDB, Pinecone, FAISS, Chroma, indexing strategies, ACID transactions, schema optimization
- **Distributed Systems:** Asynchronous workflows, concurrency handling, pessimistic locking, idempotency design, event-driven processing
- **DevOps:** Docker, Docker Compose, Railway, Render, Vercel, Hugging Face Spaces, Git
- **Core CS:** Data Structures & Algorithms, System Design, OOP, SOLID principles

PROFESSIONAL EXPERIENCE

GenAI Engineer (Full-Stack)

Aug 2025 – Present

Bosch Global Software Technologies – Hyderabad, India

- Architected multi-agent AI systems using LangChain and LangGraph with dynamic tool execution and reasoning workflows.
- Designed and deployed production-grade RAG pipelines including ingestion, chunking, embeddings, hybrid semantic retrieval, and grounded generation.
- Reduced hallucination rate by 30% through structured prompt evaluation and response validation frameworks.
- Built scalable FastAPI services with asynchronous execution and OpenAPI documentation.
- Implemented structured logging, monitoring, and containerized deployments using Docker.

Full-Stack Engineer

Jan 2025 – Aug 2025

Bosch Global Software Technologies – Hyderabad, India

- Developed modular backend services using Node.js and Spring Boot within microservices architecture.
- Optimized SQL indexing strategies reducing API latency by 20%.
- Implemented secure coding practices and CI-based deployment workflows.

Full-Stack Developer

Mar 2024 – Dec 2024

ABL Education – Noida, India

- Built full-stack React applications with RESTful backend services.
- Improved backend query performance by 25% through optimized schema design.

IntelliTrack – AI-Powered Competitive Intelligence Platform

FastAPI, PostgreSQL, OpenAI GPT-3.5, AsyncIO, Docker

- Built a SaaS-grade platform tracking competitor pricing, documentation, and changelogs.
- Implemented paragraph-level smart diffing with strategic noise filtering to eliminate dynamic content.
- Integrated LLM-based categorization of pricing, feature, and positioning changes.
- Designed snapshot versioning with historical tracking and backend health monitoring.
- Deployed full-stack solution (Frontend: Vercel, Backend: Railway).

Live: [Frontend](#) — [Backend](#) — [GitHub](#)

High-Integrity Wallet & Double-Entry Ledger System

NestJS, PostgreSQL, ACID Transactions, Concurrency Control

- Designed immutable ledger-based wallet using double-entry accounting.
- Implemented row-level pessimistic locking (SELECT FOR UPDATE) preventing race conditions.
- Enforced idempotency via unique keys guaranteeing exactly-once transaction processing.
- Ensured strict ACID compliance and transactional consistency under concurrent load.

Live: [API](#) — [GitHub](#)

High-Scale Energy Telemetry Ingestion Engine

NestJS, PostgreSQL, TypeORM, Docker

- Built ingestion system handling 14.4M records/day from 10,000+ smart meters.
- Implemented Hot/Cold data split (append-only history + upsert-based live status).
- Optimized analytics endpoint using composite indexing without full table scans.
- Designed AC/DC efficiency analytics for fleet monitoring dashboards.

Live: [API](#) — [GitHub](#)

Acme Tech RAG Chatbot (Full-Stack)

FastAPI, Pinecone, OpenAI GPT-3.5, React, Vite

- Built end-to-end RAG system with document ingestion, embedding, and semantic retrieval.
- Integrated Pinecone vector database with GPT-3.5 for grounded answer generation.
- Developed responsive React frontend deployed on Vercel.
- Designed clean API-first architecture separating ingestion and query pipelines.

Live: [Frontend](#) — [Backend](#) — [Backend Repo](#) — [Frontend Repo](#)

Production-Grade RAG Knowledge Assistant

Python, FastAPI, Vector Databases, Docker

- Built API-first Retrieval-Augmented Generation system with ingestion, chunking, embedding, and hybrid retrieval.
- Implemented source attribution and hallucination monitoring mechanisms.
- Designed modular RAG pipeline supporting document updates and incremental ingestion.
- Containerized and deployed using Docker on Hugging Face Spaces.

Live: [Hugging Face Deployment](#) — [GitHub](#)

AI Insight Dashboard – Full-Stack Analytics Platform

React, TypeScript, Node.js, REST APIs

- Developed enterprise-style analytics dashboard for structured data visualization.
- Built reusable modular UI components and optimized REST API integrations.
- Implemented dynamic charts, filtering logic, and responsive layouts.
- Deployed frontend on Vercel with production-ready configuration.

Live: [Live Demo](#) — [GitHub](#)

EDUCATION

Bachelor of Technology – Electronics and Communication Engineering

Dr. A.P.J. Abdul Kalam Technical University

2019 – 2023